Solution 1

Problem 1 — Let μ and ν be probability measures on $(E, \mathscr{E})$ and let $f : E \to [0, R]$ be a measurable function. Suppose that $\nu(A) = \mu(f\mathbb{1}(A))$ for all $A \in \mathscr{E}$. Let $(X_n : n \in \mathbb{N})$ be a sequence of independent random variables in E with law μ and let $(U_n : n \in \mathbb{N})$ be a sequence of independent random variables having the uniform law on $[0, 1]$. Set

$$T = \min\{n \in \mathbb{N} : RU_n \le f(X_n)\}, \quad Y = X_T.$$

Show that Y has law ν.

Let $\alpha = \mathbb{P}(RU_1 > f(X_1))$; note $0 \le \alpha < 1$. For any $A \in \mathscr{E}$, observe that

$$\begin{aligned}
\mathbb{P}(Y \in A) &= \sum_{n=1}^{\infty} \mathbb{P}(X_T \in A, T = n) \\
&= \sum_{n=1}^{\infty} \mathbb{P}(X_n \in A, RU_n \le f(X_n), RU_m > f(X_m) \text{ for } m < n) \\
&= \sum_{n=1}^{\infty} \mathbb{P}(X_n \in A, RU_n \le f(X_n)) \prod_{m<n} \mathbb{P}(RU_m > f(X_m)) \qquad \text{(independence)} \\
&= \sum_{n=1}^{\infty} \mathbb{P}(X_1 \in A, RU_1 \le f(X_1))\alpha^{n-1} \\
&= \frac{1}{1-\alpha}\mathbb{P}(X_1 \in A, RU_1 \le f(X_1)) \\
&= \frac{\mathbb{P}(X_1 \in A, RU_1 \le f(X_1))}{\mathbb{P}(RU_1 \le f(X_1))}
\end{aligned}$$

Cool. The numerator is just

$$\begin{aligned}
\mathbb{P}(X_1 \in A, RU_1 \le f(X_1)) &= \int_A \int_{\mathbb{R}} \mathbf{1}_{\{x : f(x) > u\}}(x, u) \, \frac{1}{R} \, du \, \mu(dx) \\
&= \int_A \int_0^{f(x)} \frac{1}{R} \, du \, \mu(dx) = \int_A \frac{f(x)}{R} \, \mu(dx) = \frac{1}{R}\nu(A)
\end{aligned}$$

and the denominator is just

$$\begin{aligned}
\mathbb{P}(RU_1 \le f(X_1)) &= \int_E \int_{\mathbb{R}} \mathbf{1}_{\{x : f(x) > u\}}(x, u) \, \frac{1}{R} \, du \, \mu(dx) \\
&= \int_E \int_0^{f(x)} \frac{1}{R} \, du \, \mu(dx) \\
&= \int_E \frac{f(x)}{R} \, \mu(dx) = \frac{1}{R} \int_E f(x) \, \mu(dx) = \frac{1}{R}
\end{aligned}$$

as desired.

We implicitly use that the event $RU_1 \le f(X_1)$ is measurable; else the question doesn't make sense. This is true, because its complement is $RU_1 > f(X_1)$, which is a countable union of measurable sets

$$\bigcup_{q \in \mathbb{Q}} \{RU_1 > q\} \cap \{f(X_1) > q\}$$

Solution 2

Problem 2 — Let $(f_n : n \in \mathbb{N})$ be a sequence of integrable functions and suppose that $f_n \to f$ a.e. for some integrable function f. Show that, if $\|f_n\|_1 \to \|f\|_1$, then $\|f_n - f\|_1 \to 0$ as $n \to \infty$.

Consider $|f_n - f| - |f_n|$. We note that

$$||f_n - f| - |f_n|| \leq |f|$$

for all n. By the dominated convergence theorem, we have

$$\begin{aligned}
\lim_{n\to\infty} \int |f_n - f| - |f_n| \, d\mu &= \int \lim_{n\to\infty} |f_n - f| - |f_n| \, d\mu \\
&= \int 0 - |f| \, d\mu \\
&= -\|f\|_1
\end{aligned}$$

but this quantity is also supposed to be

$$\begin{aligned}
\lim_{n\to\infty} \int |f_n - f| - |f_n| \, d\mu &= \lim_{n\to\infty} \int |f_n - f| \, d\mu - \lim_{n\to\infty} \int |f_n| \, d\mu \\
&= \lim_{n\to\infty} \|f_n - f\|_1 - \lim_{n\to\infty} \|f_n\|_1 \\
&= \lim_{n\to\infty} \|f_n - f\|_1 - \|f\|_1
\end{aligned}$$

Thus, $\lim_{n\to\infty} \|f_n - f\|_1 = 0$.

Solution 3

Problem 3 — Let X be a random variable and let $1 \leq p < \infty$. Show that, if $X \in L^p(\mathbb{P})$, then $\mathbb{P}[|X| \geq \lambda] = O(\lambda^{-p})$ as $\lambda \to \infty$. Prove the identity

$$\mathbb{E}[|X|^p] = \int_0^\infty p\lambda^{p-1}\mathbb{P}[|X| \geq \lambda]\, d\lambda$$

and deduce that, for all $q > p$, if $\mathbb{P}[|X| \geq \lambda] = O(\lambda^{-q})$ as $\lambda \to \infty$, then $X \in L^p(\mathbb{P})$.

Suppose $X \in L^p(\mathbb{P})$; then $\mathbb{P}(|X| \geq \lambda) \cdot \lambda^p \leq \int |X|^p \, d\mathbb{P} = M < \infty$; thus $\mathbb{P}(|X| \geq \lambda) \leq M\lambda^{-p}$, so $\mathbb{P}(|X| \geq \lambda) = O(\lambda^{-p})$. Now, we prove the identity. We have

$$\begin{aligned}
\mathbb{E}[|X|^p] &= \int_0^\infty \mathbb{P}[|X|^p \geq \lambda^p] d(\lambda^p) \\
&= \int_0^\infty p\lambda^{p-1}\mathbb{P}[|X|^p \geq \lambda^p] d\lambda
\end{aligned}$$

Thus, if $\mathbb{P}[|X| \geq \lambda] = O(\lambda^{-q})$ as $\lambda \to \infty$, then by definition there exists constants $C, \lambda_0 > 1$ such that $\mathbb{P}[|X| \geq \lambda] \leq C\lambda^{-q}$ for all $\lambda \geq \lambda_0$. This yields:

$$\begin{aligned}
\mathbb{E}[|X|^p] &\leq \int_0^{\lambda_0} p\lambda^{p-1}\mathbb{P}\left[|X|^p \geq \lambda^p\right] \mathrm{d}\lambda + \int_{\lambda_0}^\infty p\lambda^{p-1}C\lambda^{-q}\mathrm{d}\lambda \\
&\leq \int_0^{\lambda_0} p\lambda^{p-1}\mathrm{d}\lambda + \frac{Cp}{q-p}\lambda_0^{p-q} \\
&< \infty
\end{aligned}$$

Thus $X \in L^p(\mathbb{P})$.

Solution 4

Problem 4 — Show that $\|XY\|_1 = \|X\|_1 \|Y\|_1$ for independent random variables X and Y. Show further that, if X and Y are also integrable, then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

Note that $\|X\|_1 = \mathbb{E}[|X|]$, and $|XY| = |X|\,|Y|$. Both parts follow from proposition 3.7.1 in the notes. Observe that proposition 3.7.1 stats $\mathbb{E}[f(X)f(Y)] = \mathbb{E}[f(X)]\mathbb{E}[f(Y)]$ for any *bounded* measurable function f, but this condition is superfluous; the proof cites Fubini's theorem, and Fubini's theorem does not require f to be bounded; nonnegative and/or integrable functions work just fine!

Solution 5

Problem 5 — Let $(X_n : n \in \mathbb{N})$ be an identically distributed sequence in $L^2(\mathbb{P})$. Show that $n\mathbb{P}[|X_1| > \varepsilon\sqrt{n}] \to 0$ as $n \to \infty$, for all $\varepsilon > 0$. Deduce that $n^{-1/2} \max_{k \le n} |X_k| \to 0$ in probability.

Well,

$$\begin{aligned}\mathbb{E}[|X|^2] &= \int_0^\infty \mathbb{P}[|X|^2 \ge n]\, dn \\ &= \int_0^\infty \mathbb{P}[|X| \ge \sqrt{n}]\, dn\end{aligned}$$

If this is to be finite, clearly $n\mathbb{P}[|X| \ge \sqrt{n}] \to 0$ as $n \to \infty$. Indeed, the converse is that there exist arbitrarily large n such that $\mathbb{P}[|X| \ge \sqrt{n}] \ge \frac{c}{n}$ for some $c > 0$; pick some increasing sequence of those numbers $n_1, n_2, \ldots$ such that $n_{i+1} \ge 2n_i$. Then, we write:

$$\begin{aligned}&\ge \sum_i (n_{i+1} - n_i)\frac{c}{n_{i+1}} \\ &\ge \sum_i \frac{c}{2}\end{aligned}$$

which is unbounded.

If you let $n = \varepsilon^2 m$ or something, this this immediately shows the claim.

The result we wish to show is that, for all $\varepsilon > 0$,

$$\mathbb{P}\left[\max_{k \le n} |X_k| \ge \varepsilon\sqrt{n}\right] \to 0$$

However, by union bound (that is, we add up the measures of a covering), we know this is upper-bounded by

$$\sum_{k \le n} \mathbb{P}[|X_k| \ge \varepsilon\sqrt{n}] = n\mathbb{P}[|X_1| \ge \varepsilon\sqrt{n}] \to 0$$

thus the result follows.

Solution 6

Problem 6 — Let $(E, \mathscr{E}, \mu)$ be a measure space and let $V_1 \leq V_2 \leq \cdots$ be an increasing sequence of closed subspaces of $L^2 = L^2(E, \mathscr{E}, \mu)$. For $f \in L^2$, denote by f_n the orthogonal projection of f on V_n. Show that f_n converges in L^2.

First off, I'm aware that L^2 isn't a Hilbert space (the L^2 norm is a seminorm) but we can do the usual thing of identifying functions that are equal almost everywhere; this doesn't change anything in the problem, because the L^2 norm can't distinguish between functions that are equal almost everywhere.

After that, I really don't care what Hilbert space I'm in anymore; this is just functional analysis. Recall that on Hilbert spaces, orthogonal projections are a thing; for any closed subspace $V \leq E$, for any $v \in E$, there exist a unique $u \in V$ and $w \in V^\perp$ such that $v = u + w$. The projection of v onto V is u. This u is in fact the minimizer

$$u = \arg\min_{u' \in V} \left\| v - u' \right\|^2$$

Now, here's a claim:

Claim (Projections project through projections) — Suppose $W \leq V \leq E$ are a series of closed subspaces of a Hilbert space E. Pick some $v \in E$. Let $v_\perp$ be the (unique) projection of v onto V, and let $w_\perp$ be the (unique) projection of $v_\perp$ onto W. Then $w_\perp$ is also the projection of v onto W.

Proof. Note that by definition, for all $w^* \in W$, $\langle v_\perp - w^* | v - v_\perp \rangle = 0$ because $v_\perp - w^* \in V$; thus the Pythagorean theorem states that

$$\|v_\perp - w^*\|^2 = \|v - v_\perp\|^2 + \|v_\perp - w^*\|^2$$

In particular,

$$\arg\min_{w^* \in W} \|v - w^*\|^2 = \arg\min_{w^* \in W} \|v_\perp - w^*\|^2$$

and since the minimizer is unique, we have $w_\perp = \arg\min_{w^* \in W} \|v - w^*\|^2$. □

By the lemma, the projection of f_j onto V_i is f_i for all $i \leq j$. We thus write

$$f_n = \sum_{i \leq n} f_n - f_{n-1}$$

For all $n > m$, $f_n - f_{n-1} \in V_n \leq V_m$ and is thus perpendicular to $f_m - f_{m-1}$. Thus, by the pythagorean theorem,

$$\|f_n\|^2 = \sum_{i \leq n} \|f_i - f_{i-1}\|^2$$

Now, $\|f_n\|^2 \leq \|f'\|^2 < \infty$ for all n; thus

$$\sum_{i \geq 0} \|f_i - f_{i-1}\|^2 < \infty$$

This is a Cauchy sequence, so the whole summation converges in L^2 to some f':

$$\sum_{i \leq n} f_n - f_{n-1} \to f'$$

Which in general won't be f because $\bigcup V_n$ might be strictly smaller than E.

Solution 7

> **Problem 7 —** Find a uniformly integrable sequence of random variables $(X_n : n \in \mathbb{N})$ such that both $X_n \to 0$ a.s. and $\mathbb{E}[\sup_n |X_n|] = \infty$.

All of my random variables are going to be functions of a single random variable X which is uniform on $[0, 1]$. Then, X_k will take on the value of k on $(\frac{1}{k+1}, \frac{1}{k})$ and be 0 elsewhere. This is uniformly integrable, because for any $n \in \mathbb{N}$,

$$\sup \left(\mathbb{E}(|X_k| \, \mathbb{1}_{|X_k| \geq n} : k \in \mathbb{N})\right)$$

is $\frac{1}{n+1}$ achieved by X_n; this vanishes as $n \to \infty$.

On the other hand,

$$\mathbb{E}[\sup_n |X_n|] \geq \mathbb{E}[\sup_{n \leq k} |X_n|] = \frac{1}{2} + \cdots + \frac{1}{k+1} \to \infty$$

Thus $\mathbb{E}[\sup_n |X_n|] = \infty$.

Solution 8

Problem 8 — Let $(X_n : n \in \mathbb{N})$ be an identically distributed sequence in $L^2(\mathbb{P})$. Show that $\mathbb{E}[\max_{k\le n} |X_k|]/\sqrt{n} \to 0$ as $n \to \infty$.

Let the internals be $Y_n = n^{-1/2} \max_{k\le n} |X_k|$; we wish to show $\mathbb{E}[Y_n] \to 0$. Note that

$$\mathbb{E}[|Y_N|^2] = \mathbb{E}[\frac{1}{n} \max_{k\le n} |X_k|^2] \le \frac{1}{n} \sum_{k\le n} \mathbb{E}[|X_k|^2] = \mathbb{E}[|X_1|^2]$$

By Holder's (Cauchy-Schwarz) inequality, $\mathbb{E}[|Y_N| \mathbb{1}_A] \le \|X\|_2 \mathbb{P}(A)^{1/2}$ which is finite, thus $\mathbb{E}[|Y_n| \mathbb{1}_A] \to 0$ uniformly as $\mu(A) \to 0$, i.e. Y_n is UI. We also know $Y_n \to 0$ in probability from problem 5, so by theorem 6.2.3 we know $Y_n \to 0$ in L^1, which is exactly what we want.